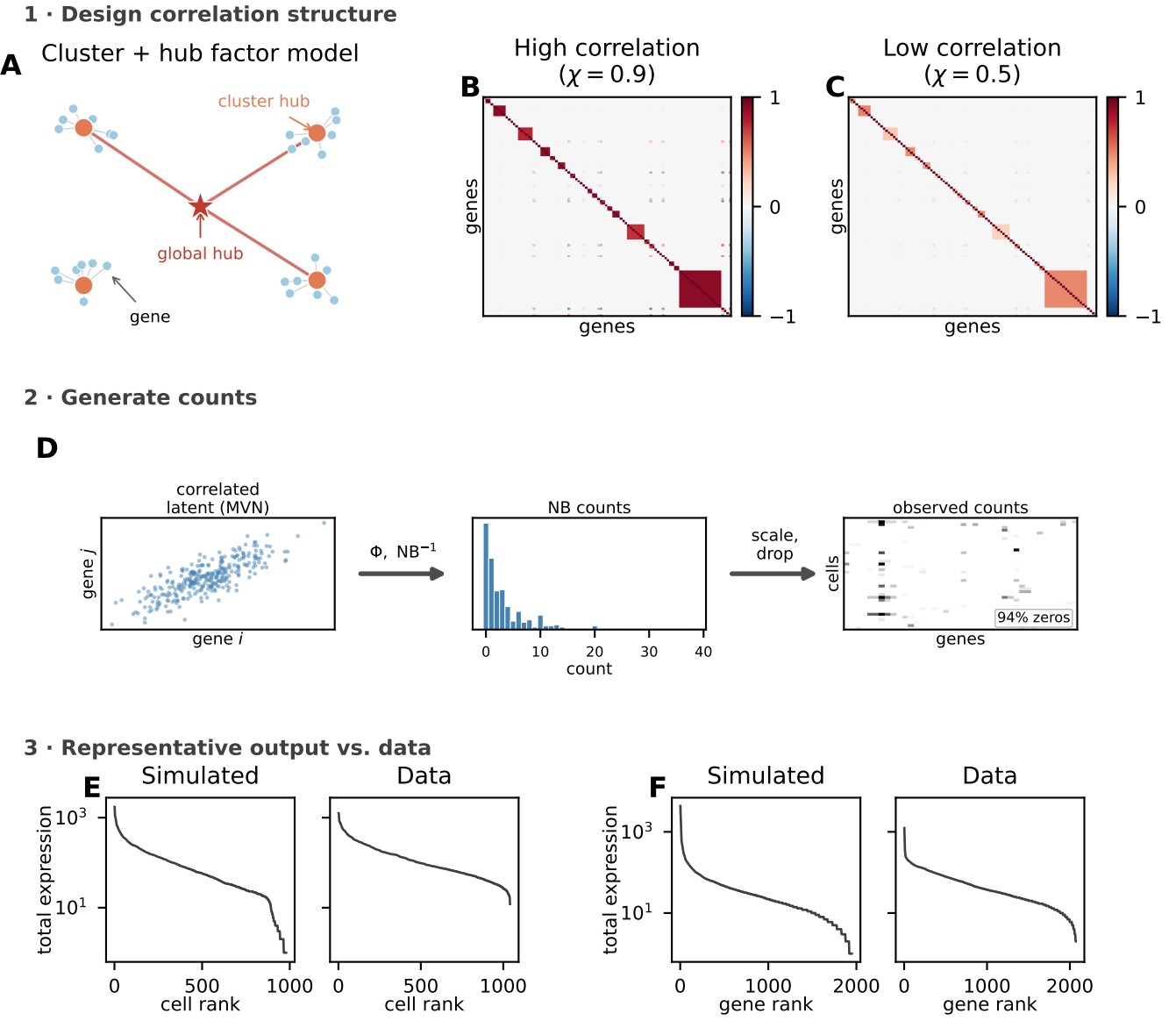

1 · Design correlation structure
A
Cluster + hub factor model
cluster hub
global hub
gene
B
High correlation
($\chi = 0.9$)
genes
genes
1
0
−1
C
Low correlation
($\chi = 0.5$)
genes
genes
1
0
−1
2 · Generate counts
D
correlated
latent (MVN)
gene j
gene i
Φ, NB^{-1}
NB counts
0
10
20
30
40
count
scale,
drop
observed counts
cells
94% zeros
genes
3 · Representative output vs. data
E
Simulated
Data
total expression
10^3
10^1
0
500
1000
cell rank
0
500
1000
cell rank
F
Simulated
Data
total expression
10^3
10^1
0
1000
2000
gene rank
0
1000
2000
gene rank